Ziyu (Rick) Xu

Robotics Institute, Carnegie Mellon University

Research Interests: Robot Foundation Models, Real-World Policy Adaptation, and Learning-Augmented Control

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Education

Carnegie Mellon University (CMU) **Aug. 2026**
Master of Science in Robotics (MSR) Pittsburgh, PA

University of Michigan, Ann Arbor (UM) **Oct. 2024 – May 2026**
B.S.E. in Mechanical Engineering (Robotics Track); Minor in Computer Science GPA: 3.85/4.0 (Top 5%)

- **Core Coursework:** Robot Learning, RL, Embedded Systems, Optimal Control, Convex Optimization

Shanghai Jiao Tong University (SJTU) **Sep. 2022 – Jul. 2026**
B.S.E. in Electrical and Computer Engineering GPA: 3.56/4.0 (Top 20%)

- **Core Coursework:** Algorithms, Mathematical Foundations, Probability Theory, Signals and Systems

Research Experiences and Internships

Unitree Robotics, R&D Department, Embodied AI Team **May 2026 – Present**
Research Intern, Humanoid Robot Foundation Models Hangzhou, China

- ***Closed-Loop Real-Robot Infrastructure for Humanoid Foundation-Model Policies:*** Built and validated an end-to-end policy infrastructure on Unitree H2/G1 humanoids, integrating teleoperation data collection, VLA/WAM and whole-body policy adapters, rollout logging and evaluation, controller interfaces, and policy fine-tuning. Enabled launch with one command and closed-loop rollouts across platforms, allowing policies and controllers to be integrated without modifying platform-specific launch workflows.
- ***Embodiment VLA Adaptation & Post-Training:*** Conducted a structured survey of representative robot VLA, WAM, and physical-AI policies, especially work from NVIDIA GEAR Lab. Deployed and fine-tuned Isaac GR00T/OpenPI VLA policies with SONIC whole-body control on Unitree humanoids using real-robot rollout traces. Currently developing a phase-aware adaptation method that aligns semantic task progress and execution timing across embodiments for policy transfer and rollout-driven post-training.
- ***Reproducible Real-World VLA Benchmark for Humanoid Manipulation:*** Implemented an 8-task, 24-scene benchmark with green-screen background replacement to evaluate foundation models under varied visual conditions. Built a PICO-based scene-digitization and AR-guided reconstruction pipeline that captured shelf-object layouts as reusable digital scene templates, and registered target placements back into the physical workspace, achieving 2.5 cm median placement error.

Shanghai Artificial Intelligence Lab & IRMV, SJTU **Apr. 2026 – Present**
Research Assistant, Directed by Prof. Junchi Yan and Prof. Hesheng Wang Shanghai, China

- ***Action-Step-Aware Dynamic Visual Sparsity for VLAs:*** Developing a fine-grained, action-step-aware visual-token pruning method that re-estimates token importance and retention budgets across action-generation steps rather than pruning once at prefill, preserving task-relevant visual evidence as manipulation context evolves, and preparing a first-author manuscript on efficient vision-language-action models.

Computational Autonomy and Robotics Lab, UM **Jun. 2025 – Apr. 2026**
Research Assistant, Directed by Prof. Maani Ghaffari Ann Arbor, MI

- ***Online Learning Lie-MPC for Robust ASV Control:*** Developed a Neural-Fly-inspired learning-augmented Lie-MPC framework for an autonomous surface vehicle, estimating residual wind/current disturbances online and injecting learned dynamics corrections into the controller for robust trajectory tracking under hydrodynamic uncertainty. Implemented and evaluated the framework on a real ASV, achieved a 20% improvement in trajectory-tracking performance over the baseline MPC controller during field experiments.
- ***Field Perception for Aquatic Vegetation Tracking:*** Built a water-scene perception pipeline for autonomous surface vehicles, combining DINO/SAM2-based segmentation with Grounded-DINO and YOLO fine-tuning to detect submerged vegetation and provide perception signals for navigation and autonomy experiments.
- ***ASV Hardware and Sensor Integration for Closed-Loop Field Autonomy:*** Designed sensor mounts and integrated GPS, lidar, IMU, and camera systems on an autonomous surface vehicle. Aligned and calibrated sensors to support field data collection and control experiments.

Intelligent Robotics and Autonomy Lab , UM **Oct. 2024 – Apr. 2026**
Research Assistant, Directed by Prof. Vasileios Tzoumas Ann Arbor, MI

- ***Multi-Robot Active-SLAM Simulation & ROS2 Integration:*** Rebuilt a Unity + AirSim multi-drone simulation testbed with redesigned vehicle dynamics and sensing modules, then migrated SlideSLAM communication logics from ROS1 to ROS2 by refactoring topic interfaces and workflows for bandwidth-limited active-SLAM evaluation.

- ***Resource-Aware Multi-Robot Exploration Algorithms:*** Developed distributed exploration EXP3-style algorithms for bandwidth-limited robot teams, formulating communication-aware action selection with submodular objectives and bandit feedback in simulation to balance map coverage and communication cost.

Surgical Computer Vision Detection, SJTU **Apr. 2024 – Dec. 2024**
Research Assistant, Directed by Prof. Yutong Ban Shanghai, China

- ***Foundation Segmentation Model Evaluation for Surgical Video:*** Evaluated Segment Anything Model variants on surgical video frames, analyzing domain shift and temporal mask stability around instruments, tissues, and tool-tissue interaction regions.

Publications and Manuscripts

- **Ziyu Xu**, Haoran Zhang, and Sen Peng.
"PhaseVLA: Semantic Phase and Speed Alignment for Cross-Embodiment VLA Models",
Manuscript in preparation, 2026.
- **Ziyu Xu**, Xiaohan Wang, Anning Hu, Hesheng Wang, and Junchi Yan.
"Action-Step-Aware Dynamic Visual Sparsity for Efficient Vision-Language-Action Models",
Manuscript in preparation, 2026.
- Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari.
"Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control",
IEEE Robotics and Automation Letters (RA-L), revised manuscript under review, 2025.
Contribution: Led the design and development of the learning-augmented Lie-MPC framework; contributed to real-ASV experiments and evaluation.
- C. Yuan, J. Jiang, K. Yang, **Z. Xu**, . . . , and Y. Ban.
"Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis", *npj Digital Surgery*, accepted, 2024.

Additional Experience

- **RL-Control led Multi-Branch Cooling System for Data center, Grundfos Company:** Designed an RL control algorithm for a multi-branch cooling system with heterogeneous thermal loads, regulating individual water-pump speeds to minimize energy consumption by 30% relative to the provided trim-response algorithm.
- **Low-Cost Automated 96-Well Plate Washer:** Designed, built, and delivered a fully automated 96-well microplate washer to the University of Michigan Medical School at 10% of the cost of a comparable commercial system.
- **Data Engineering & Web Systems, Library of Shanghai Jiao Tong University:** Led a team of 4 to build a Python dataset and internal web platform supporting 4,000+ SJTU faculty.
- **Quantitative Analysis, APEXUS Tech LLC:** Developed a hybrid statistical-learning and factor-mining model, integrating it into an execution framework with reproducible experiment logs and adaptive parameter tuning.

Awards

- Gold Medal in 2023 University Physics Competition, by American Physical Society & American Astronomical Society (top 1%)
- Third place in 2023 RoboMaster Campus Competition (top 10%)
- Dean's Honor List, University of Michigan College of Engineering (2024, 2025)
- M Prize in 2024 Mathematical Contest in Modeling (top 13%)

Skills

- **Robot Learning & Control:** VLA post-training, humanoid foundation models, whole-body policy, RL, real-robot policy deployment, teleoperation/data collection, MPC, sim-to-real
- **Perception & Multimodal ML:** PyTorch, OpenCV, SAM2, Grounded-DINO, YOLO, DINO features, visual token compression, spatial grounding, LLMs fine-tuning
- **Systems & Tools:** ROS/ROS2, Python/C++, Unity + AirSim, Docker, Linux, Git, sensor integration, CAD, FEA, Manufacturing, embedded system